Non-monotone circuit lower bounds from efficient monotone simulations

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Allowed standard prerequisites (editorial).

- Known monotone lower bounds, logspace algorithms and threshold circuit bounds. Exact author prior theorem statements retained with citations. Full author depth-reduction/padding support proofs included.

3 Preliminaries

3.1 Notation

Boolean functions. We denote by Mono the set of all monotone Boolean functions. We define $\mathsf{poly} = \{n \mapsto n^C : C \in \mathbb{N}\}$. A Boolean function $f : \{0,1\}^{\binom{n}{2}} \to \{0,1\}$ is said to be a graph property if $f(G) = f(H)$ for any two isomorphic graphs G and H. Let $\mathcal{F} = \{f_n\}_{n\in\mathbb{N}}$ be a sequence of graph properties, where f_n is defined over undirected graphs on n vertices. We say that $\mathcal{F}$ is *preserved under homomorphisms* if, whenever there is a homomorphism from a graph G to a graph H, we have $\mathcal{F}(G) \leq \mathcal{F}(H)$. We denote by $\mathsf{HomPreserving}$ the set of all graph properties which are preserved under homomorphisms. Note that $\mathsf{HomPreserving} \subseteq \mathsf{Mono}$.

Boolean circuits. We denote by $\mathsf{AC}^0_d[s]$ the family of Boolean functions computed by size-s, depth-d Boolean circuits with unbounded fan-in $\{\wedge, \vee\}$-gates and input literals from $\{x_1, \overline{x_1}, \ldots, x_n, \overline{x_n}\}$. We write $\mathsf{AC}^0[s]$ as a shorthand for $\bigcup_{d=1}^{\infty} \mathsf{AC}^0_d[s]$, and AC^0 as a shorthand of $\mathsf{AC}^0[n^{O(1)}] = \mathsf{AC}^0[\mathsf{poly}]$. We will also refer to $\mathsf{AC}^0_d[\mathsf{poly}]$ by AC^0_d. We write $\mathsf{DNF}[s]$ to denote the family of Boolean functions computed by size-s DNFs, where size is measured by number of terms. We write $\mathsf{CNF}[s]$ analogously. We write $\mathsf{SIZE}[s]$ to denote the family of Boolean functions computed by size-s circuits. We write $\mathsf{DEPTH}[d]$ to denote the family of Boolean functions computed by fan-in 2 circuits of depth d. We denote by $\mathsf{AC}^0[\oplus]$ the family of Boolean functions computed by polynomial-size AC^0 circuits with unbounded fan-in $\oplus$-gates.

We denote by L the family of Boolean functions computed by logspace machines, and by NL the family of Boolean functions computed by polynomial-time nondeterministic logspace machines. Moreover, we denote by $\oplus\mathsf{L}$ the family of Boolean functions computed by polynomial-time nondeterministic logspace machines with a *parity* acceptance condition (i.e., an input is accepted if the number of accepting paths is odd).

Circuit complexity. Given a circuit class $\mathcal{C}$, we write $\mathsf{m}C$ to denote the monotone version of $\mathcal{C}$. Given a function f, we write $\mathsf{mSIZE}(f)$ to denote the size of the smallest monotone circuit computing f and $\mathsf{mDEPTH}(f)$ to denote the smallest depth of a fan-in 2 monotone circuit computing f. Given two Boolean functions f, g, we write $f \leq_m^{\mathsf{mProj}} g$ if there exists a many-one reduction from f to g in which each bit of the reduction is a monotone projection[1] of the input.

Miscellanea. Let $\alpha \in \{0,1\}^n$. We define $|\alpha|_1 := \sum_{i=1}^n \alpha_i$. We call $|\alpha|_1$ the *Hamming weight* of α. We let $\mathrm{supp}(\alpha) = \{i \in [n] : \alpha_i = 1\}$. We let $\mathrm{THR}_{k,n} : \{0,1\}^n \to \{0,1\}$ be the Boolean function such that $\mathrm{THR}_{k,n}(x) = 1 \iff |x|_1 \geq k$.

3.2 Background results

The next lemma, which is proved via a standard "guess-and-verify" approach, shows that nondeterministic logspace computations can be simulated by circuits of size 2^{n^ε} and of depth $d = O_\varepsilon(1)$.

Lemma 3.1 (Folklore; see, e.g., [AHM+08, Lemma 8.1])**.** *For all $\varepsilon > 0$, we have $\mathsf{NL} \subseteq \mathsf{AC}^0[2^{n^\varepsilon}]$.*

Theorem 3.2 (Polynomial-size constant-depth vs. larger monotone depth)**.** *For every $k \geq 1$, we have $\mathsf{AC}^0 \cap \mathsf{Mono} \not\subseteq \mathsf{mDEPTH}[(\log n)^k]$. Moreover, this separation holds for a monotone graph property.*

Theorem 3.3 (Polynomial-size constant-depth vs. larger monotone size)**.** *For every $k \geq 1$, we have $\mathsf{AC}^0[\oplus] \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{(\log n)^k}]$.*

Theorem 3.4 ([GKRS19])**.** *There exists $\varepsilon > 0$ such that $\oplus\mathsf{L} \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{o(n^\varepsilon)}]$. Moreover, this separation is witnessed by* 3-XOR-SAT*.*

Theorem 3.5. *For every $i \geq 1$, we have $\mathsf{AC}^0 \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \not\subseteq \mathsf{mDEPTH}[(\log n)^i]$. In particular, we have $\mathsf{AC}^0 \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \not\subseteq \mathsf{mNC}^i$.*

Theorem 3.6 (New circuit lower bounds from monotone simulations)**.** *There exists $\varepsilon > 0$ such that the following holds.*

1. *If $\mathsf{AC}^0_3 \cap \mathsf{Mono} \subseteq \mathsf{mNC}^1$, then $\mathsf{NP} \not\subseteq \mathsf{AC}^0_3[2^{o(n)}]$.*
2. *If $\mathsf{AC}^0_4 \cap \mathsf{Mono} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$, then $\mathsf{NP} \not\subseteq \mathsf{AC}^0_4[2^{o(\sqrt{n}/\log n)}]$.*
3. *If $\mathsf{AC}^0 \cap \mathsf{Mono} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$, then $\mathsf{NC}^2 \not\subseteq \mathsf{NC}^1$.*
4. *If $\mathsf{NC}^1 \cap \mathsf{Mono} \subseteq \mathsf{mSIZE}[2^{O(n^\varepsilon)}]$, then $\mathsf{NC}^2 \not\subseteq \mathsf{NC}^1$.*
5. *If $\mathsf{AC}^0 \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$, then $\mathsf{NP} \not\subseteq \mathsf{NC}^1$.*
6. *If $\mathsf{NC}^1 \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$, then $\mathsf{L} \not\subseteq \mathsf{NC}^1$.*

Lemma 3.7 (Folklore; see, e.g., [AKR+01, OSS19])**.** *Let $f : \{0,1\}^n \to \{0,1\}$ be a Boolean function computed by a $\oplus\mathsf{L}$ machine. For every $\delta > 0$, there exists an $\mathsf{AC}^0[\oplus]$ circuit of size 2^{n^δ} that computes f.*

Proof. Let M be a $\oplus$L-machine computing f. Without loss of generality, we may assume that each configuration in the configuration graph G of M is *time-stamped* – in other words, each configuration carries the information of the number of computational steps it takes to arrive at it.[2] We may also assume that every accepting computation takes exactly the same amount of time, which means that every path from the starting configuration v_{start} to the accepting configuration v_{accept} has the same length in the configuration graph. These assumptions imply that the configuration graph is *layered* (because a configuration with time-stamp t can only point to configurations with time-stamp $t+1$) and acyclic. Note that, for a fixed machine, the configuration graph can be computed from the input string using a projection.

[1] A monotone projection is a projection without negations.

[2] Formally, we can define a $\oplus$L-machine M' such that the configurations of M' are (C, t), where C is a configuration of M, and $t = 0, 1, \ldots, m = n^{O(1)}$ is a number denoting the time in which the configuration was achieved. A configuration (C, t) can only reach a configuration $(C', t+1)$ in the configuration graph of M'.

Let $m = n^{O(1)}$ be the time that an accepting computation takes. We now show how to count (modulo 2) the number of accepting paths from v_{start} to v_{accept} with a depth-d $\mathsf{AC}^0[\oplus]$ circuit. First, choose $m^{1/d}-1$ configurations $v_1, \ldots, v_{m^{1/d}-1}$ (henceforth called "checkpoints") from $V(G)$, such that the configuration v_i is at the level $i \cdot m^{1-1/d}$ in the configuration graph (i.e., it takes $i \cdot m^{1-1/d}$ time steps to arrive at v_i). For convenience, we let $v_0 = v_{\mathsf{start}}$ and $v_{m^{1/d}} = v_{\mathsf{accept}}$. We then count the number of paths from from v_{start} to v_{accept} that go through $v_1, \ldots, v_{m^{1/d}-1}$, and sum over all possible choices of the checkpoints. Since the graph is layered and each path from v_0 to $v_{m^{1/d}}$ has length exactly m, there is only one choice of checkpoints that witnesses a given path from v_0 to $v_{m^{1/d}}$, so no path is counted twice in this summation. Letting $\#\mathsf{paths}(s, t, \ell)$ denote the number of paths between configurations s and t with distance exactly ℓ, we obtain

$$\#\mathsf{paths}(v_0, v_{m^{1/d}}, m) = \sum_{v_1,\ldots,v_{m^{1/d}-1}} \prod_{i=0}^{m^{1/d}-1} \#\mathsf{paths}(v_i, v_{i+1}, m^{1-1/d}).$$

The above calculation can be done in modulo 2 with an unbounded fan-in XOR gate (replacing the summation) and an unbounded fan-in AND gate (replacing the product). Note that the formula above is recursive. Repeating the same computation for calculating (modulo 2) the expression $\#\mathsf{paths}(v_i, v_{i+1}, m^{1-1/d})$ for each i, we obtain a depth-$2d$ $\mathsf{AC}^0[\oplus]$ circuit for calculating the number of paths from v_{start} to v_{accept} (modulo 2). Clearly, the total size of the circuit is $2^{O(m^{1/d} \cdot \log m)}$, which is smaller than 2^{n^δ} for a large enough constant d. □

First, we recall a result of [BGW99], which proves monotone lower bounds for the following function. Let $\mathsf{OddFactor}_n : \{0,1\}^{\binom{n}{2}} \to \{0,1\}$ be the function that accepts a given graph if it contains an *odd factor* – in other words, a spanning subgraph in which the degree of every vertex is odd. Babai, Gál and Wigderson [BGW99] proved the following result:

Theorem 3.8 ([BGW99])**.** *Any monotone formula computing* $\mathsf{OddFactor}_n$ *has size* $2^{\Omega(n)}$*, and any monotone circuit computing* $\mathsf{OddFactor}_n$ *has size* $n^{\Omega(\log n)}$*.*

The proof in [BGW99] is actually for the case of *bipartite graphs*, but it easily extends to general graphs, since the bipartite case reduces to the general case by a monotone projection. The formula lower bound stated above is slightly stronger because it makes use of asymptotically optimal lower bounds on the randomized communication complexity of DISJ_n [KS92], which were not available to [BGW99]. We remark that, with a different language, a monotone circuit lower bound for $\mathsf{OddFactor}$ is also implicitly proved in Feder and Vardi [FV98, Theorem 30].

We now recall an upper bound for $\mathsf{OddFactor}$, implicitly proved in an unpublished note due to Johannsen [Joh03].

Theorem 3.9 ([Joh03])**.** *We have* $\mathsf{OddFactor} \in \mathsf{L}$*.*

Proof. We first recall the following observation about the $\mathsf{OddFactor}$ function, which appears in different forms in the literature (see [Urq87, Lemma 4.1] or [Juk12, Lemma 18.16]; see also [Joh03, Proposition 1] for a different proof.)

Claim. *A graph G has an odd factor if and only if every connected component of G has an even number of vertices.*

Proof. If a graph G has an odd factor, we can conclude that every connected component of G has an even number of vertices from the well-known observation that in every graph there is an even number of vertices of odd degree.

Now suppose that every connected component of G has an even number of vertices. We will iteratively construct an odd factor F of G. We begin with the empty graph. We take any two vertices u, v in the same connected component of G which currently have even degree in F, and consider any path $P = (x_1, \ldots, x_k)$ between u and v, where $x_1 = u$ and $x_k = v$. If the edge $x_i x_{i+1}$ is currently in F, we remove $x_i x_{i+1}$ from F; otherwise, we add $x_i x_{i+1}$ to F. It's easy to check that, in every iteration of this procedure, only the vertices u and v have the parity of their degree changed in F; the degree of every other vertex stays the same (modulo 2). Since every connected component has an even number of vertices, this means that, eventually, every vertex in F will have odd degree. □

Now it's easy to check in logspace if every connected component of G has an even number of vertices using Reingold's algorithm for undirected connectivity [Rei05]. It suffices to check if, for every vertex v of G, the number of vertices reachable from v is odd. □

Now, if we only desire to obtain a function in AC^0 not computed by monotone circuits of depth $(\log n)^i$, we can follow the same argument of Theorem 3.3, using Lemma 3.1 instead of Lemma 3.7. In order to obtain moreover a monotone *graph property* witnessing this separation, we will need the following lemma, which enables us to obtain a graph property after "padding" a graph property. We defer the proof of this lemma to the end of this section.

Lemma 3.10. *Let $f : \{0,1\}^{\binom{n}{2}} \to \{0,1\}$ be a monotone graph property on graphs of n vertices. The following holds.*

1. *If $f \in \mathsf{NC}^i$ for some $i > 1$, then there exists a monotone graph property g on graphs of $N = 2^{(\log n)^i}$ vertices such that $g \in \mathsf{NC}^1$ and $f \leq_m^{\mathsf{mProj}} g$.*
2. *If $f \in \mathsf{NL}$, then for all $\varepsilon > 0$ there exists a monotone graph property g on graphs of $N = 2^{n^\varepsilon}$ vertices such that g can be computed by AC^0 circuits of size $N^{2+o(1)}$ and $f \leq_m^{\mathsf{mProj}} g$.*
3. *If $f \in \oplus\mathsf{L}$, then for all $\varepsilon > 0$ there exists a monotone graph property g on graphs of $N = 2^{n^\varepsilon}$ vertices such that g can be computed by $\mathsf{AC}^0[\oplus]$ circuits of size $N^{2+o(1)}$ and $f \leq_m^{\mathsf{mProj}} g$.*

We are now ready to prove Theorem 3.5.

Proof of Theorem 3.5. Fix $n \in \mathbb{N}$ and take an $\varepsilon < 1/i$. Observing that $\mathsf{L} \subseteq \mathsf{NL}$, from Theorem 3.9 and item (2) of Lemma 3.10 we conclude that there exists a monotone graph property f on $N = 2^{n^\varepsilon}$ vertices such that $f \in \mathsf{AC}^0$ and $\mathsf{OddFactor}_n \leq_m^{\mathsf{mProj}} f$. By Theorem 3.8, any monotone circuit computing f has depth $\Omega(n) = \Omega((\log N)^{1/\varepsilon}) \gg (\log N)^i$. □

Raz and Wigderson [RW92] observed that there exists a monotone function $f \in \mathsf{NC}^1 \setminus \mathsf{mNC}$. Using Lemma 3.10, we observe moreover that it's possible to obtain this separation with a monotone graph property.

Proposition 3.11. *We have $\mathsf{NC}^1 \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \not\subseteq \mathsf{mNC}$.*

Proof. Observing that $\mathsf{L} \subseteq \mathsf{NC}^2$, we conclude from Theorem 3.9 and item (1) of Lemma 3.10 that there exists a monotone graph property f on $N = 2^{(\log n)^2}$ vertices such that $f \in \mathsf{NC}^1$ and $\mathsf{OddFactor}_n \leq_m^{\mathsf{mProj}} f$. By Theorem 3.8, any monotone circuit computing f has depth $\Omega(n) = \Omega(2^{\sqrt{\log N}})$, which implies $f \notin \mathsf{mNC}$. □

3.3 Efficient monotone padding for graph properties

We will now prove Lemma 3.10. We first recall some low-depth circuits for computing threshold functions, which we will use to design a circuit for efficiently computing the adjacency matrix of induced subgraphs.

Theorem 3.12 ([HWWY94])**.** *Let $d > 0$ be a constant. The function $\mathrm{THR}_{(\log n)^d, n}$ can be computed by an AC^0 circuit of size $n^{o(1)}$ and depth $d + O(1)$.*

Theorem 3.13 ([AKS83])**.** *For every $k \in [n]$, the function $\mathrm{THR}_{k,n}$ can be computed by a circuit of depth $O(\log n)$ and size $n^{O(1)}$.*

Lemma 3.14. *There exists a circuit C_n^k with $\binom{n}{2} + n$ inputs and $\binom{k}{2}$ outputs which, when given as input an adjacency matrix of a graph G on n vertices and a characteristic vector of a set $S \subseteq [n]$ such that $|S| \leq k$, outputs the adjacency matrix of the graph $G[S]$, padded with isolated vertices when $|S| < k$. The circuit has constant-depth and size $n^{2+o(1)}$ when $k = \mathrm{polylog}(n)$, and size $n^{O(1)}$ and depth $O(\log n)$ otherwise.*

Proof. Let $\{x_{ij}\}_{i,j \in [n]}$ encode the adjacency matrix of G. Let $\alpha \in \{0,1\}^n$ be the characteristic vector of S. Let $i, j \in [k]$. Note that $\{i, j\} \in E(G[S])$ if and only if there exists $a, b \in [n]$ such that

- α_a is the i-th non-zero entry of α,
- α_b is the j-th non-zero entry of α, and
- $x_{ab} = 1$ (i.e., a and b are connected in G).

We first consider the case $k = \text{polylog}(n)$. In this case, the first two conditions can be checked with circuits of size $n^{o(1)}$ using Theorem 3.12. Therefore, we can compute if i and j are adjacent using $n^{2+o(1)}$ gates and constant depth. As there are at most $(\log n)^{O(1)}$ such pairs, we can output $G[S]$ with at most $n^{2+o(1)}$ gates.

For any k, the first two conditions can be checked with an NC^1 circuit by Theorem 3.13. Since there are at most n^2 pairs i, j, the entire adjacency matrix can be computed with a $O(\log n)$-depth and polynomial-size circuit. □

We are ready to prove Lemma 3.10.

Proof of Lemma 3.10. We first prove (1). Fix $n \in \mathbb{N}$ and let $N = 2^{(\log n)^i}$. For a graph G on N vertices such that $|E(G)| \leq \binom{n}{2}$, let G_{clean} be the graph obtained from G by removing isolated vertices from G one-by-one, in lexicographic order, until one of the following two conditions are satisfied: (1) there are no more isolated vertices in G_{clean}, *or* (2) G_{clean} has exactly n vertices. Let $g : \{0,1\}^{\binom{N}{2}} \to \{0,1\}$ be the monotone graph property defined as follows:

$$g(G) := \left(|E(G)| > \binom{n}{2}\right) \vee (|V(G_{\mathsf{clean}})| > n) \vee (f(G_{\mathsf{clean}}) = 1).$$

Note that g accepts a graph G if and only if at least one of the following three conditions are satisfied:

1. G has at most $\binom{n}{2}$ edges, G_{clean} has exactly n vertices and $f(G_{\mathsf{clean}}) = 1$, or
2. G has more than $\binom{n}{2}$ edges, or
3. G_{clean} has more than n vertices.

We observe that the monotonicity of g follows from the monotonicity of f. We also claim that g is a graph property. Indeed, the graph G_{clean} is the same (up to isomorphism), irrespective of the order according to which the isolated vertices are removed from G. Moreover, the function f is also a graph property. Because of this, all the three conditions above are preserved under isomorphisms.

We first observe that f is a monotone projection of g. Indeed, given a graph G on n vertices, we can easily construct by a monotone projection a graph G' on N vertices and at most $\binom{n}{2}$ edges such that $f(G) = g(G')$. We just let G' have a planted copy of G, and all other vertices are isolated. Then $G'_{\mathsf{clean}} = G$ (up to isomorphism) and $g(G') = f(G_{\mathsf{clean}}) = f(G)$.

We now show how to compute g in NC^1. Let $\{x_{ij}\}_{i,j\in[N]}$ be the input bits of g, corresponding to the adjacency matrix of a graph G. The circuit computes as follows.

1. If $|E(G)| > \binom{n}{2}$, accept the graph G.
2. Compute the characteristic vector $\alpha \in \{0,1\}^N$ of the set of all non-isolated vertices of G. If $|\alpha|_1 > n$, accept the graph G.
3. Compute G_{clean} and output $f(G_{\mathsf{clean}})$.

Note that checking if $|E(G)| > \binom{n}{2}$ can be done in NC^1 by Theorem 3.13. Moreover, for all $i \in [N]$, we have $\alpha_i = \bigvee_{j\in[N]} x_{ij}$, and therefore α_i can be computed by a circuit of depth $O(\log N)$ and $O(N)$ gates. In total, the vector α can be computed with $O(N^2)$ gates and $O(\log N)$ depth. Finally, we can check if $|\alpha|_1 > n$ in NC^1 with a threshold circuit.

For the final step, we compute G_{clean}. If $|\alpha|_1 = n$, note that $G_{\mathsf{clean}} = G[\text{supp}(\alpha)]$. When $|\alpha|_1 < n$, then G_{clean} is $G[\text{supp}(\alpha)]$ padded with isolated vertices. We can therefore compute G_{clean} with the circuit C^n_N of Lemma 3.14. Moreover, since $f \in \mathsf{NC}^i$, we have that f can be computed by a circuit of size $n^{O(1)} = N^{o(1)}$ and depth $O((\log n)^i) = O(\log N)$. Therefore, computing $f(G_{\mathsf{clean}})$ can be done in NC^1. Overall, we get that $g \in \mathsf{NC}^1$.

In order to prove (2), it suffices to modify the proof above. The modification can be briefly described as follows. We let $N = 2^{n^\varepsilon}$. Every time Lemma 3.14 is applied, we use the AC^0 circuit instead of the NC^1 circuit, since $n = \text{polylog}(N)$. This ammounts to $N^{2+o(1)}$ many gates with unbounded fan-in. Moreover, since by assumption $f \in \mathsf{NL}$, applying Lemma 3.1 we obtain an AC^0 circuit for f of size $2^{n^{\varepsilon/2}} = N^{o(1)}$, so we can compute $f(G_{\mathsf{clean}})$ in constant depth with $N^{o(1)}$ gates.

Finally, for (3) it suffices to apply the same argument used for (2), replacing an application of Lemma 3.1 by an application of Lemma 3.7. □

3.4 Non-monotone lower bounds from monotone simulations

We now show that if monotone circuits are able to efficiently simulate non-monotone circuits computing monotone Boolean functions, then striking complexity separations follow. We also show a result of this kind for simulations of graph properties. We first prove a lemma connecting the simulation of AC^0 circuits with the simulation of NL machines.

Lemma 3.15. *For all constants $\varepsilon > 0$ and $C \geq 1$, if $\mathsf{AC}^0 \cap \mathsf{Mono} \subseteq \mathsf{mSIZE}[2^{O((\log n)^C)}]$, then $\mathsf{NL} \cap \mathsf{Mono} \subseteq \mathsf{mSIZE}[2^{o(n^\varepsilon)}]$.*

Proof. We prove the contrapositive. Suppose that there exists $\varepsilon > 0$ such that $\mathsf{NL} \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{o(n^\varepsilon)}]$. This means that there exists a monotone function f such that $f \in \mathsf{NL}$ and any monotone circuit computing f has size $2^{\Omega(n^\varepsilon)}$.

Let $\delta = \varepsilon/(2C)$ and let $m = 2^{n^\delta}$. Let $g : \{0,1\}^n \times \{0,1\}^m \to \{0,1\}$ be the Boolean function defined as $g(x,y) = f(x)$. Note that g is a function on $N := m + n = 2^{\Theta(n^\delta)}$ bits. By Lemma 3.1, there exists an AC^0 circuit computing f of size $2^{n^\delta} = N^{O(1)}$. Moreover, any monotone circuit computing g has size $2^{\Omega(n^\varepsilon)} = 2^{\Omega((\log N)^{\varepsilon/\delta})} = 2^{\Omega((\log N)^{2C})}$. □

Next, we recall the strongest known monotone circuit and formula lower bounds for a monotone function in NP.

Theorem 3.16 ([PR17])**.** $\mathsf{NP} \cap \mathsf{Mono} \not\subseteq \mathsf{mDEPTH}[o(n)]$.

Theorem 3.17 ([CKR20])**.** $\mathsf{NP} \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{o(\sqrt{n}/\log n)}]$.

We are now ready to state and prove our first result regarding new complexity separations from monotone simulations. Recall that obtaining explicit lower bounds against depth-3 AC^0 circuits of size $2^{\omega(n^{1/2})}$ is a major challenge in circuit complexity theory, while the best lower bound on the size of depth-4 AC^0 circuits computing a function in NP is currently $2^{\Omega(n^{1/3})}$ [Hås86]. Moreover, no strict separation is known in the following sequence of inclusions of complexity classes: $\mathsf{ACC} \subseteq \mathsf{TC}^0 \subseteq \mathsf{NC}^1 \subseteq \mathsf{L} \subseteq \mathsf{NL} \subseteq \oplus\mathsf{L} \subseteq \mathsf{NC}^2$. We show that efficient monotone simulations would bring new results in both of these fronts. (We stress that all lower bound consequences appearing below refer to separations against non-uniform circuits.)[3]

Theorem 3.18. *Let $\mathcal{C}$ be a class of circuits. There exists $\varepsilon > 0$ such that the following holds:*

1. *If $\mathsf{AC}^0_3 \cap \mathsf{Mono} \subseteq \mathsf{mNC}^1$, then $\mathsf{NP} \not\subseteq \mathsf{AC}^0_3[2^{o(n)}]$.*
2. *If $\mathsf{AC}^0_4 \cap \mathsf{Mono} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$, then $\mathsf{NP} \not\subseteq \mathsf{AC}^0_4[2^{o(\sqrt{n}/\log n)}]$.*
3. *If $\mathcal{C} \cap \mathsf{Mono} \subseteq \mathsf{mSIZE}[2^{O(n^\varepsilon)}]$, then $\mathsf{NC}^2 \not\subseteq \mathcal{C}$.*
4. *If $\mathsf{AC}^0 \cap \mathsf{Mono} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$, then $\mathsf{NC}^2 \not\subseteq \mathsf{NC}^1$.*

Proof. We will prove each item separately.

Proof of (1). Let us assume that $\mathsf{AC}^0_3 \cap \mathsf{Mono} \subseteq \mathsf{mNC}^1$. Let f be the function of Theorem 3.16. For a contradiction, suppose that $f \in \mathsf{AC}^0_3[2^{o(n)}]$. Let $\alpha : \mathbb{N} \to \mathbb{N}$ be such that $\alpha(n) \to_n \infty$ and f has a depth-3 AC^0 circuit of size $2^{n/\alpha}$. Let $m = 2^{n/(10\cdot\alpha)}$ and let $g : \{0,1\}^n \times \{0,1\}^m \to$

[3] In other words, all upper bounds are *uniform*, but the lower bounds hold even for *non-uniform* circuits. Note that this is stronger than lower bounds for uniform circuits.

$\{0,1\}$ be the function $g(x,y) = f(x)$. Let $N = n + m = (1 + o(1))2^{n/(10\cdot\alpha)}$. Clearly, the function g has a depth-3 AC^0 circuit of size $2^{n/\alpha} = N^{O(1)}$. Since g is monotone, we conclude from the assumption that g is computed by a polynomial-size monotone formula. Now, since $f(x) = g(x, 1^m)$, we obtain a monotone formula of size $N^{O(1)} = 2^{o(n)}$ for computing f, which contradicts the lower bound of Theorem 3.16.

Proof of (2). Similar to the proof of item (1), but using Theorem 3.17 instead.

Proof of (3). Suppose that $\mathsf{NC}^2 \subseteq \mathcal{C}$. By Theorem 3.4, there exists a monotone function $f \in \mathsf{NC}^2$ on n bits and a number $\varepsilon > 0$ such that $f \notin \mathsf{mSIZE}[2^{o(n^\varepsilon)}]$. Therefore, for any $\delta > 0$ such that $\delta < \varepsilon$, we have $f \notin \mathsf{mSIZE}[2^{O(n^\delta)}]$. Since, by assumption, we have $f \in \mathsf{NC}^2 \subseteq \mathcal{C}$, we obtain $\mathcal{C} \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{O(n^\delta)}]$.

Proof of (4). If $\mathsf{NC}^2 \subseteq \mathsf{NC}^1$, then, by item (3), we get $\mathsf{NC}^1 \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{o(n^\varepsilon)}]$. From Lemma 3.15, we obtain $\mathsf{AC}^0 \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[\mathsf{poly}]$. □

As a motivation to the ensuing discussion, we recall a result of Rossman [Ros08a], who showed that any homomorphism-preserving graph property computed by AC^0 circuits is also computed by monotone AC^0 circuits.

Theorem 3.19 ([Ros08a])**.** $\mathsf{AC}^0 \cap \mathsf{HomPreserving} \subseteq \mathsf{mDNF}[\mathsf{poly}]$.

This inspires the question of whether general graph properties can also be efficiently simulated by monotone circuits. We show that, if true, such simulations would imply strong complexity separations. Let us first recall an exponential monotone circuit lower bound for monotone graph properties, and we will be ready to state and prove our main result.

Theorem 3.20 ([AB87])**.** *There exists $\varepsilon > 0$ such that* $\mathsf{NP} \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \not\subseteq \mathsf{mSIZE}[2^{o(n^\varepsilon)}]$.

Theorem 3.21. *Let $\mathcal{C}$ be a class of circuits. The following holds:*

1. *If $\mathcal{C} \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$, then $\mathsf{L} \not\subseteq \mathcal{C}$.*
2. *If $\mathcal{C} \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \subseteq \mathsf{mDEPTH}[o(\sqrt{n})]$, where n denotes the number of input bits, then $\mathsf{L} \not\subseteq \mathcal{C}$.*
3. *If $\mathsf{AC}^0 \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$, then $\mathsf{NP} \not\subseteq \mathsf{NC}^1$.*

Proof. We will prove each item separately.

Proof of (1). Suppose that $\mathsf{L} \subseteq \mathcal{C}$. By Theorem 3.8, the monotone graph property $\mathsf{OddFactor}$ satisfies $\mathsf{OddFactor} \notin \mathsf{mSIZE}[\mathsf{poly}]$. Moreover, we have the upper bound $\mathsf{OddFactor} \in \mathsf{L}$ by Theorem 3.9. Since, by assumption, we have $\mathsf{OddFactor} \in \mathsf{L} \subseteq \mathcal{C}$, we obtain $\mathcal{C} \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \not\subseteq \mathsf{mSIZE}[\mathsf{poly}]$.

Proof of (2). Suppose that $\mathsf{L} \subseteq \mathcal{C}$. By Theorems 3.8 and 3.9, there exists a monotone graph property $f \in \mathsf{L}$ such that $f \notin \mathsf{mDEPTH}[o(\sqrt{n})]$. Since, by assumption, we have $f \in \mathsf{L} \subseteq \mathcal{C}$, we obtain $\mathcal{C} \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \not\subseteq \mathsf{mDEPTH}[o(\sqrt{n})]$.

Proof of (3). Suppose that $\mathsf{NP} \subseteq \mathsf{NC}^1$. By Theorem 3.20, there exists a monotone graph property $f \in \mathsf{NC}^1$ such that $\mathsf{mSIZE}(f) = 2^{\Omega(n^\varepsilon)}$ for some $\varepsilon > 0$. Let $\delta = \varepsilon/2$. By Lemma 3.10 (Item 2), there exists a monotone graph property g on $N = 2^{n^\delta}$ vertices computed by an AC^0 circuit of size $N^{2+o(1)}$ such that f is a monotone projection of g. Theorem 3.20 implies that any monotone circuit computing f has size $2^{\Omega(n^\varepsilon)} = 2^{\Omega((\log N)^2)} = N^{\omega(1)}$. □

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